

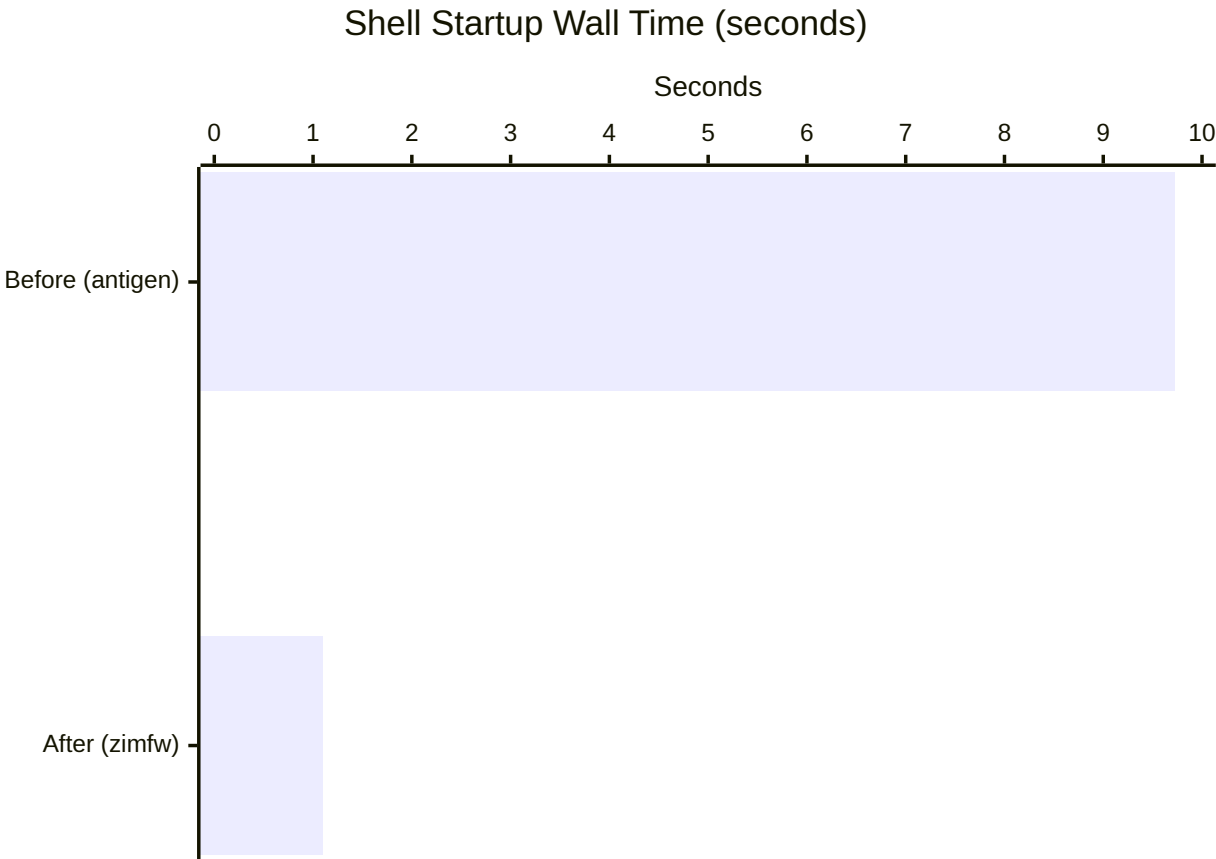
Shell Startup Improvement — zimfw Migration

Date: 2026-06-13

Change: Replaced antigen with zimfw as the zsh plugin manager via `CLI_CONFIG_MODULES=zimfw`

Wall Time

	Before	After	Improvement
Wall time	9.727s	1.097s	~9× faster
User time	0.41s	0.36s	
System time	0.56s	0.43s	

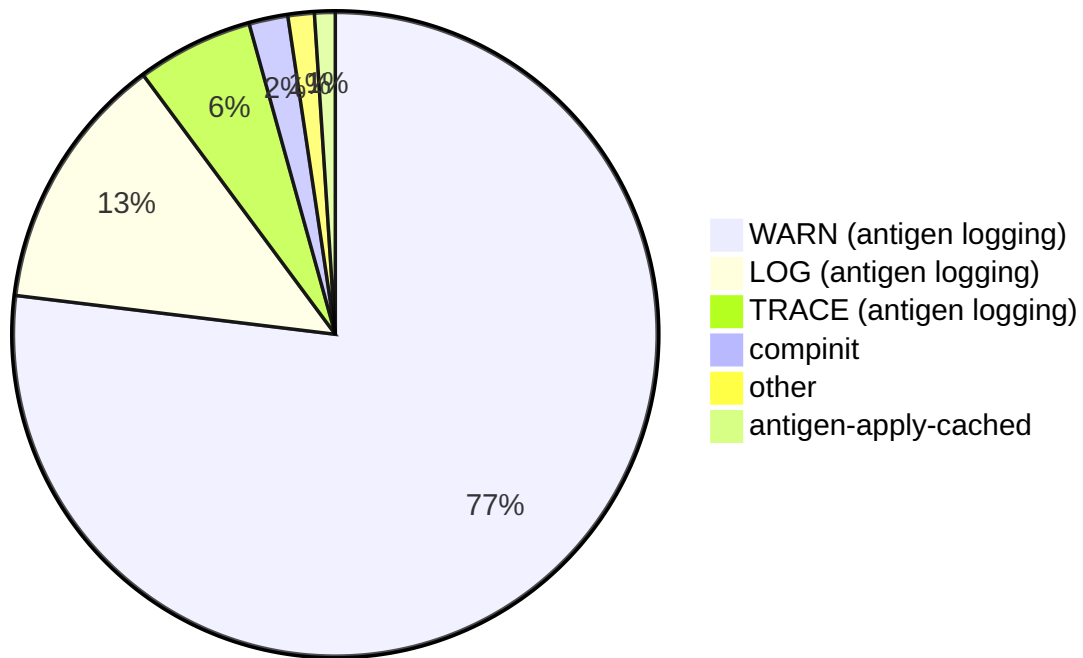


Where Time Was Spent

Before — antigen (9.7s total tracked)

Antigen’s internal WARN/LOG/TRACE logging dominated. Every shell start triggered runtime plugin resolution.

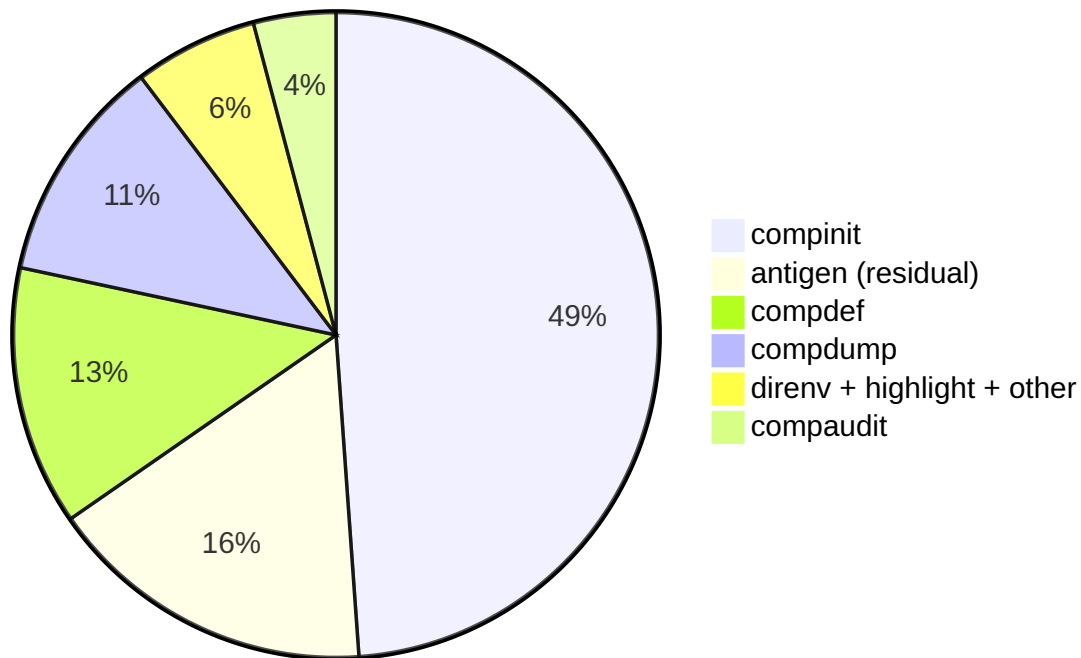
zprof self time — Before (antigen)



After — zimfw (1.1s total tracked)

Antigen logging noise is gone. compinit is now the dominant cost — previously invisible under antigen overhead.

zprof self time — After (zimfw)

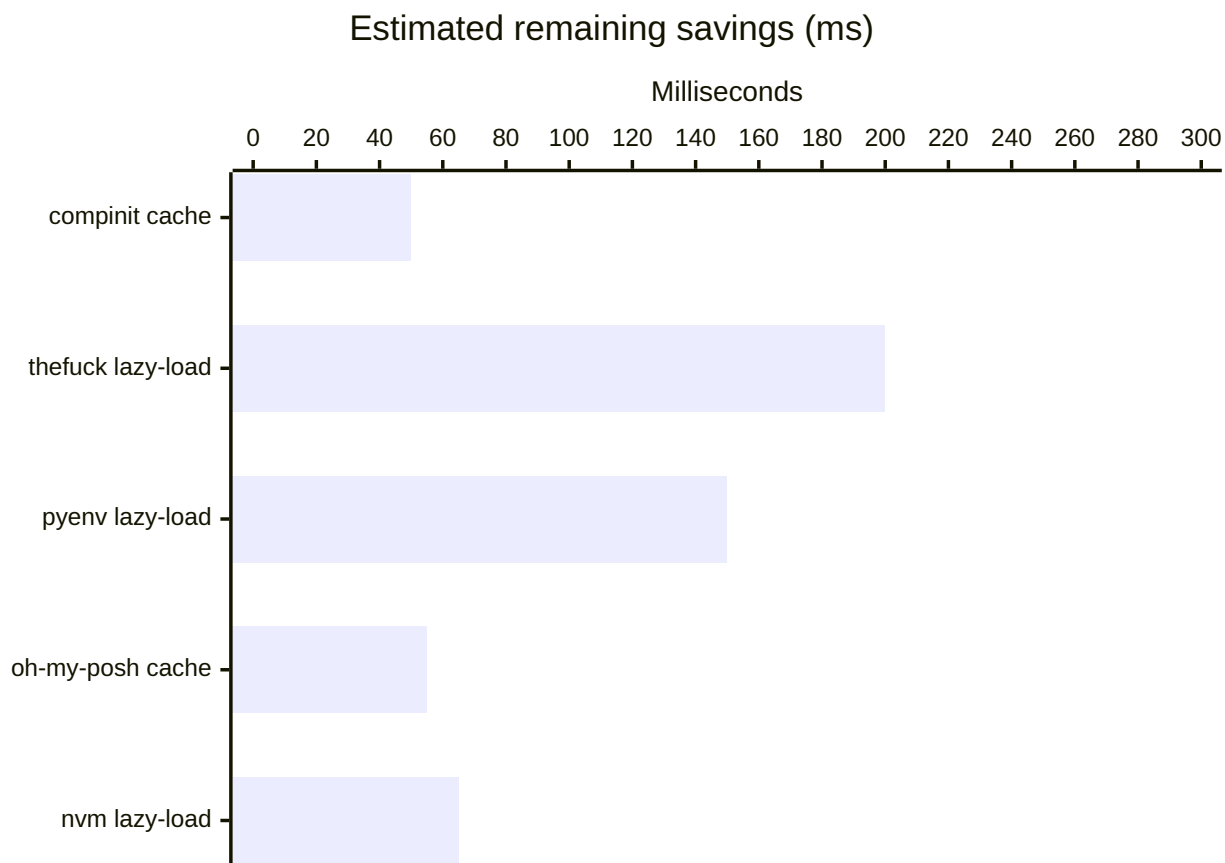


Why zimfw Is Faster

Antigen resolves plugins at runtime on every shell start — cloning, caching, and applying 6 plugins through a hook-based pipeline. Its internal WARN/LOG/TRACE calls alone accounted for **93%** of all tracked function time.

zimfw generates a static `init.zsh` at install time (`cli-config install -t zimfw`). Shell startup just **sources** that file — no git operations, no runtime resolution, no logging overhead.

Remaining Opportunities



Fix	Est. saving	Status
Cache <code>compinit</code> (skip rebuild if < 24h old)	20–50ms	Pending
Lazy-load <code>thefuck</code>	100–300ms	Pending
Lazy-load <code>pyenv</code> <code>init</code>	100–200ms	Pending
Cache <code>oh-my-posh</code> <code>init</code> output	30–80ms	Pending
Re-enable <code>nvm</code> lazy-load	30–100ms	Pending
Native <code>CLI_CONFIG_ROOT</code> detection	2–10ms	Pending
Remove <code>find/rm .zwc</code> from default <code>.zshrc</code>	5–20ms	Pending

See `plans-startup-improvement.md` for implementation details on each.